

Finite-Center Accountability and the Positive Route to Binary Rank

Live review note of June 20, 2026

Abstract

Two finite-center obstruction notes show that quotient-visible operational, sector, modular, recovery, continuum, or entropy data do not by themselves reconstruct hidden finite central rank or the predicate $q = 2$. This short note states the matching positive criterion. For a supplied finite center $\mathcal{Z}_F \simeq \ell^\infty(F)$, let $V_F = \mathcal{Z}_F/\mathcal{C}1$ be its contrast space. A declared family of probes, effects, outcome statistics, sector labels, recovery tasks, modular records, finite-alphabet laws, or direct measurements generates a readout map $R : V_F \rightarrow W$. The supplied data determine exactly the quotient $V_F/\ker R$. Hence, in a comparison class where the hidden finite center may vary and q is not supplied as an external label, full finite central cardinality is recoverable from the readout data only when R is faithful on the physical finite-center contrast space and an admissibility or maximality premise says this represented center is the whole physical finite center. Under those premises $q = \text{rank } R + 1$. A binary conclusion $q = 2$ additionally needs one accounted contrast plus a theorem or measurement identifying that contrast as the full finite center rather than a quotient of a larger hidden extension.

This is deliberately demotion-ready. In finite statistical language it is a Blackwell/sufficiency statement. In operator-algebraic language it is a Petz-style relative-to-the-supplied-channel boundary. The possible value is only the translation into black-hole/AQFT reconstruction: recovery, entropy, sector, modular, or locally covariant records force q or $q = 2$ only after finite-center accountability has been supplied.

Status. This is live non-frozen review metadata. It does not modify either frozen Team 1 paper, Lean archive, public mirror manifest, or SHA-256 package. Its purpose is external classification: known theorem, false statement, missing hypothesis, too broad physically, or useful expository corollary.

1 Question

Positive reconstruction theorems reconstruct the object supplied to them: a represented von Neumann algebra, a locally covariant functor, a sector category, a correctable code

algebra, a channel tested by specified preparations and effects, or a Type II algebra with trace and normalization data. The finite-center obstruction is the converse question. If the displayed data have already passed through a quotient or coarse readout, what extra premise is needed before hidden finite central rank, and in particular the binary predicate $q = 2$, becomes reconstructible?

The word “reconstructible” is used throughout in the fiber sense: a target must be a function of the displayed readout data on the stated comparison class. There is no obstruction if q , the full finite center, or a complete atomic measurement is simply supplied as part of the input. The obstruction is only to recovering that target from records that have already forgotten some finite central directions.

2 Finite Accountability

Definition 1 (finite-center readout family). Let F be a finite set with $q = |F|$, and let

$$\mathcal{Z}_F = \ell^\infty(F), \quad V_F = \mathcal{Z}_F / \mathbb{C}1.$$

Thus $\dim V_F = q - 1$. A finite-center readout for F is a linear map

$$R_F : V_F \longrightarrow W_F$$

generated by the declared observation family: normal probes, effects, outcome statistics, sector records, recovery tasks, modular records, a finite-alphabet law, or a direct measurement channel. Its null directions are

$$N_F = \ker R_F.$$

The comparison class may contain many finite centers F , many readout maps R_F , or hidden refinements of a represented quotient. The target $q = |F|$ is reconstructed only if it is a function of the displayed data on this whole comparison class.

Theorem 1 (finite accountability quotient). *Let $\mathcal{C}$ be a comparison class of finite-center readout data (F, R_F, D_F) , where D_F depends on finite-center contrasts only through $R_F : V_F \rightarrow W_F$. Assume $\mathcal{C}$ allows hidden finite centers to be refined, quotiented, or extended in directions not seen by the displayed readout, and that $q = |F|$ is not separately supplied as an input label. Then:*

1. *For each F , D_F factors through V_F / N_F .*
2. *The displayed data reconstruct the accounted finite-center contrast rank $\dim(V_F / N_F) = \text{rank } R_F$, not the unaccounted hidden cardinality.*
3. *Full finite central cardinality q is reconstructible on $\mathcal{C}$ only after adding both $N_F = 0$ and an admissibility or maximality premise saying that the represented finite center is the whole physical finite center, not a quotient, null sector, gauge reduction, excluded sector, or hidden extension.*

4. Under those added premises, for each object of $\mathcal{C}$,

$$q = \dim(V_F/N_F) + 1 = \text{rank } R_F + 1.$$

Proof. For a fixed F , if two contrasts $v, v' \in V_F$ have the same image under R_F , then $v - v' \in N_F$. Every record built from R_F assigns the same value to v and v' , so it factors through V_F/N_F . Consequently any target that changes within a coset of N_F is not a function of the supplied data.

The finite commutative algebra $\ell^\infty(F)$ has one atomic coordinate for each element of F . After constants are quotiented out, its contrast space has dimension $q - 1$. If $N_F = 0$, the readout is faithful on that contrast space, so it determines $\dim V_F$. If the admissibility premise also says the represented finite center is physically exhaustive, this dimension is the physical finite-center contrast dimension, and $q = \dim V_F + 1$. If $N_F \neq 0$, or if the represented center is only a quotient or admissible subcenter of a larger hidden center, refinements inside the null or hidden directions change the unaccounted finite cardinality while preserving all R_F -visible data. Hence, over $\mathcal{C}$, the data determine accounted rank but not hidden cardinality unless the faithfulness and exhaustion premises are added. $\square$

Remark 1 (supplied labels). The theorem does not deny that a laboratory, model definition, or microscopic postulate may hand over q directly. In that case q is an input label, not a reconstruction from quotient-visible readout data. The Team 1 boundary concerns black-hole/AQFT arguments where entropy, recovery, sector, modular, or locally covariant records are used to infer a finite alphabet that has not already been supplied.

Corollary 1 (binary rank requires two inputs). *A conclusion $q = 2$ from finite-center readout data requires both:*

$$\dim(V_F/N_F) = 1$$

and a maximality, admissibility, direct-measurement, or finite-alphabet theorem identifying that accounted one-dimensional contrast as the full physical finite center. One accounted contrast alone proves only that the declared experiment sees a binary quotient.

3 Relation to Standard Sufficiency

The theorem should be demoted if a standard source already states this exact translation. Blackwell comparison of experiments makes sufficiency relative to a declared experiment or decision problem [1, 2]; Le Cam's deficiency theory gives a broader comparison-of-experiments language for approximate reconstruction loss [3, 4]. Petz sufficiency makes recovery and equality of relative entropy relative to a declared subalgebra or channel [5, 6], and quantum comparison theorems extend the same relative-to-a-model boundary to finite quantum statistical models and completely positive maps [7, 8]. Von Neumann algebra ideal theory says ultraweakly closed two-sided ideals are central-projection summands [9, 10]. None of these facts is new.

The possible finite-center contribution is the reconstruction boundary: black-hole/AQFT records determine a finite central cardinality only when their hypotheses

include faithful finite-center accountability. Without that premise, quotient-visible entropy, recovery, sector, modular, or locally covariant records reconstruct the supplied quotient, not hidden cardinality.

Internal classification after sufficiency pass. After checking Blackwell comparison, Le Cam deficiency, Petz sufficiency, and Buscemi–Shmaya quantum comparison results, Team 1 should classify the mathematical core of Theorem 1 as known and demoted. The quotient calculation is the target-specific sufficiency/factorization condition for a declared experiment, channel, subalgebra, or statistical model. The only remaining candidate value is the black-hole/AQFT translation: entropy, recovery, sector, modular, or locally covariant records do not force a finite alphabet or binary rank unless those records are sufficient for that exact finite-center target.

4 Claim Boundary for Reviewers

Claim	Status
R determines $V_F/\ker R$.	Elementary finite-dimensional linear algebra.
R determines full q when $\ker R = 0$ and the represented center is physically exhaustive.	Positive accountability criterion.
One accounted central contrast implies $q = 2$.	False without maximality, admissibility, direct measurement, or a finite-alphabet theorem.
Blackwell/Le Cam/Petz/quantum comparison theory may already contain the criterion.	Reviewer should identify the exact theorem if so; Team 1 should demote.
AQFT/OAQEC hypotheses may already supply maximality or faithful support.	Reviewer should name the hypothesis and decide whether it is present in the Team 1 statement.

5 Reviewer Question

Is Theorem 1 merely finite Blackwell/Le Cam/Petz or quantum-statistical comparison theory under a different name, or does the black-hole/AQFT translation isolate a useful claim boundary? If it is standard, the desired response is the citation and the exact theorem name. If it is false or too broad, the desired response is a counterexample or missing physical hypothesis.

References

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